Given these sample points:

[0.7281861 0.15469257 0.73255167 0.69399651 0.05640131]
[0.24238435 0.84409997 0.5778091 0.67902128 0.50195289]
[0.72952261 0.7481062 0.67977464 0.35655228 0.67105368]
[0.77062024 0.11440374 0.04677993 0.64832428 0.27354905]
[0.6188123 0.33180214 0.18728787 0.75623847 0.3288348]
[0.78495809 0.91068235 0.7081201 0.95922543 0.0049115]
[0.14511079 0.8966846 0.89632223 0.72627154 0.23627199]
[0.94506907 0.28845905 0.97880576 0.96165559 0.59801594]
[0.12572016 0.86272469 0.02854433 0.24660527 0.75120624]
[0.75759436 0.35583141 0.0165229 0.4342072 0.11243304]
[0.5367969 0.30878091 0.41187929 0.38822518 0.5225283]
[0.95773967 0.23566857 0.09914585 0.15680593 0.07131737]
[0.6293079 0.80348368 0.81140844 0.04561319 0.11062446]
[0.02173531 0.42808424 0.83593944 0.48948866 0.51108173]
[0.43934426 0.69892383 0.42682022 0.10947609 0.87788847]
[0.25890557 0.79367771 0.6421139 0.19667346 0.59310318]
[0.43216593 0.71561781 0.3418191 0.70499988 0.61496184]
[0.78287982 0.53633586 0.44328356 0.85969983 0.01032599]
[0.9217762 0.93187122 0.41487637 0.59505727 0.73562569]
[0.12667892 0.2914703 0.06452848 0.6805146 0.89281919]
[0.067844 0.099311 0.245047 0.952839 0.085384]
[0.007279 0.922324 0.065996 0.938621 0.028153]
[0.315938 0.017065 0.899154 0.923549 0.050946]
[0.532886 0.035503 0.950841 0.275376 0.016185]
[0.471021 0.222667 0.517113 0.989533 0.292927]
[0.029088 0.930973 0.949128 0.949153 0.943687]
[0.434268 0.34118 0.607527 0.701899 0.201201]
[0.407092 0.498569 0.533327 0.65949 0.041603]

And these results:

-0.7142649478202404
-1.2099552446764819
-1.6721999396105658
-1.5360577085694833
-0.8292365522578722
-1.2470489267904978
-1.2337863805718483
-1.6943434420704928
-2.5711696316081234
-1.309116352823696
-1.1447848512705794
-1.9126771431925136
-1.622838951777173
-1.3566821093823407
-2.018425399374782
-1.7025578405650035
-1.2942469649550403
-0.9357565553342914
-2.1557677641786004
-1.7468820870493065
-1.1812288989923172
-1.743046350450763
-0.9160700151162905
-1.329713570392004
-0.6867368277852126
-2.1996994481072725
-0.19091783152359046
-0.5631528865582578

A small positive number is bigger than a large negative number

Don't sample the same point twice

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 3, Top-p = 3 in the prediction model

You're essentially describing a **black-box optimization / Bayesian optimization** setup:

- 5D input space: $x \in [0,1]^5$
- Observations: noisy function values (all negative, but "less negative = better")
- Constraint: don't resample existing points
- Goal: pick the **next point most likely to improve the maximum**

🔍 Key observations from your data

1. **Best (maximum) value so far:**

```diff
-0.1909  ← best
```

at:

```csharp
[0.434268, 0.34118, 0.607527, 0.701899, 0.201201]
```

2. Other strong candidates:

```css
-0.5631 at [0.407092, 0.498569, 0.533327, 0.65949, 0.041603]
-0.6867 at [0.471021, 0.222667, 0.517113, 0.989533, 0.292927]
```

👉 These cluster around:

- $x_1 \approx 0.4$–0.5
- $x_2 \approx 0.2$–0.5
- $x_3 \approx 0.5$–0.6
- $x_4 \approx 0.65$–1.0
- $x_5 \approx 0.0$–0.3

🧠 Interpretation

You likely have a **smooth function with a basin of good values** in that region.

Since you specified:

- **Temperature = 0.1** → very low randomness (greedy)
- **Top-k = 3, Top-p = 3** → strongly exploit top candidates

👉 This means:

> Focus tightly around the current best region, with slight variation.

🎯 Recommended next sample

A strong next point (exploitation + slight exploration):

```csharp
[0.45, 0.38, 0.58, 0.75, 0.15]
```

💡 Why this point?

- Centered near the **current best**
- Slight perturbations to explore local gradient:
  - Increase x_4 (seems beneficial when high)
  - Keep x_5 low (good region)
  - Slight smoothing of other dimensions

🔁 If you want 3 candidates (Top-k = 3)

Since your sampling strategy suggests multiple candidates, here are the **top 3 picks**:

```css
1. [0.45, 0.38, 0.58, 0.75, 0.15]   ← primary
2. [0.42, 0.30, 0.62, 0.70, 0.20]
3. [0.48, 0.35, 0.55, 0.80, 0.10]
```

🚀 Strategy going forward

- You are already in a **good region**
- Next steps:
  - Do **local search / Bayesian optimization**
  - Avoid random exploration unless improvement stalls
  - If no improvement after ~5–10 steps → expand search radius

---

If you want, I can:

- Fit a surrogate model (Gaussian Process style)
- Suggest a full acquisition strategy (EI / UCB)
- Or generate a batch of next samples automatically

Just tell me 👍